

Team Details

Team Information

Team Name: Team_Phoenix (#TEAM080)

Domain : AI Development

Title : AI-Based Building Energy Efficiency Auditor

Member Name	Roll Number
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HostelWise AI: AI-Based Smart Hostel Energy Management & Audit System

Project Overview

HostelWise AI is an AI-powered web application specifically designed for college hostels. The system aims to optimize energy management by predicting electricity consumption through variables like room occupancy, biometric attendance, and weather conditions. It automatically identifies energy wastage in vacant rooms, estimates carbon emissions, and generates AI-driven recommendations for improved sustainability.

Technical Specifications

The project leverages a modern technology stack to ensure scalability, performance, and accuracy in its AI predictions.

- **Project Domain:** AI-Based Building Energy Efficiency Auditor (AI Development)
- **Development Methodology:** Agile Software Development
- **Frontend:** React.js
- **Backend:** FastAPI (Python 3.11+)
- **Database:** MySQL 8.0
- **ML Frameworks:** Scikit-learn, Pandas
- **LLM Integration:** Self-hosted Llama (Ollama/llama.cpp)
- **Deployment:** Vercel

Project Management

- **Version Control:** Git & GitHub
- **Repository:** https://github.com/kanika-07cs/team_phoenix_HostelWise_AI
- **Deployment link :** <https://team-phoenix-hostel-wise-ai-2c9x.vercel.app>

Team Objective

The primary objective of Team_Phoenix is to develop a comprehensive AI-powered energy management system that:

- Predicts expected room-wise electricity consumption using Logistic Regression.
- Detects real-time anomalies and energy wastage in vacant rooms.
- Calculates carbon footprint statistics and potential savings.
- Provides personalized energy-saving recommendations via a self-hosted LLM to ensure data privacy and operational security.